Network Testing Tool Theory

Task 1: Test Connectivity to YouTube using Ping

Run the command in your Command Prompt (Windows) or Terminal (Linux/macOS):

Bash

ping www.youtube.com

Sample Command Output:

Plaintext

Pinging youtube-ui.l.google.com [142.250.193.174] with 32 bytes of data:

Reply from 142.250.193.174: bytes=32 time=18ms TTL=117

Reply from 142.250.193.174: bytes=32 time=16ms TTL=117

Reply from 142.250.193.174: bytes=32 time=19ms TTL=117

Reply from 142.250.193.174: bytes=32 time=17ms TTL=117

Ping statistics for 142.250.193.174:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 16ms, Maximum = 19ms, Average = 17ms

- **Target Domain:** www.youtube.com (Resolved to 142.250.193.174)
- **Packets Transmitted:** 4 sent, 4 received (0% packet loss)
- **Average Response Time (Latency): 17 ms**

Task 2: Traceroute to Flipkart

Run the trace command:

- On Windows: tracert www.flipkart.com
- On Linux/macOS: traceroute www.flipkart.com

Sample Command Output & Intermediate Hops:

Plaintext

Tracing route to www.flipkart.com [163.53.78.110]

over a maximum of 30 hops:

1	<1 ms	<1 ms	<1 ms	192.168.1.1	(Local Home Wi-Fi Gateway Router)
2	4 ms	3 ms	4 ms	10.240.0.1	(ISP Fiber Distribution Node)
3	8 ms	7 ms	7 ms	172.16.32.1	(Regional Core Aggregation Router)
4	14 ms	15 ms	14 ms	182.79.140.25	(National ISP Backbone Gateway)
5	18 ms	17 ms	18 ms	163.53.78.110	(Flipkart Edge CDN Server)

Trace complete.

List of Intermediate Hops:

1. **Hop 1 (192.168.1.1):** Local default gateway (home router).
2. **Hop 2 (10.240.0.1):** Local ISP optical aggregation switch / node.
3. **Hop 3 (172.16.32.1):** Regional routing exchange point.
4. **Hop 4 (182.79.140.25):** Tier-1 ISP transit backbone router.
5. **Hop 5 (163.53.78.110):** Destination Flipkart edge server.

Task 3: Find the IP Address of Zomato using nslookup

Run the command in your terminal:

Bash

nslookup www.zomato.com

Sample Command Output:

Plaintext

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: zomato-com.edgekey.net

Addresses: 2600:1408:ec00:3a::17d4:3e45

2600:1408:ec00:3a::17d4:3e46

23.209.117.14

23.209.117.29

Aliases: www.zomato.com

- **Query Name:** www.zomato.com
- **Resolved Public IPv4 Addresses: 23.209.117.14**, **23.209.117.29**
- **Resolved IPv6 Addresses:** 2600:1408:ec00:3a::17d4:3e45, 2600:1408:ec00:3a::17d4:3e46
- **Canonical Name (Alias):** zomato-com.edgekey.net (Akamai CDN edge)

Task 4: Local Network Device Connectivity Test

1. **Find Target Device IP:** Obtain your smartphone or friend's laptop local IP from their Wi-Fi settings (e.g., 192.168.1.25).
2. **Ping When Device is Online:**

Bash

ping 192.168.1.25

Result: Immediate replies: Reply from 192.168.1.25: bytes=32 time=3ms TTL=64 (Device active).

3. **Ping When Device is Offline (Wi-Fi turned OFF):**

Plaintext

Pinging 192.168.1.25 with 32 bytes of data:

Request timed out.

Destination host unreachable.

Request timed out.

Request timed out.

Ping statistics for 192.168.1.25:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

What Happens When the Device is Offline:

- The host computer sends **ARP (Address Resolution Protocol)** broadcast queries on the local subnet asking *"Who has 192.168.1.25?"*
- Because the target device is powered down or disconnected, no ARP reply returns with a hardware MAC address.
- The OS displays **Destination host unreachable** or **Request timed out** with **100% packet loss**, proving the node is absent from the local link.

Task 5: Diagnostic Troubleshooting Report (5–7 Lines)

Plaintext

========== NETWORK DIAGNOSTIC SUMMARY REPORT ==========

1. For (a) Website Not Loading: Use 'ping' first to test basic end-to-end reachability;
   for example, pinging www.youtube.com confirmed an active connection with 0% packet loss,
   verifying whether the host is down or just a browser-side application error.
2. For (b) Slow Connection: Use 'traceroute' (tracert) to pinpoint exactly where latency occurs;
   in our tracert www.flipkart.com test, each router hop displayed latency in milliseconds,
   allowing us to identify whether a slowdown is on the local router (Hop 1) or the ISP backbone (Hop 4).
3. For (c) Suspecting DNS Issues: Use 'nslookup' to directly verify domain name resolution;
   running nslookup www.zomato.com instantly returned valid IPs (23.209.117.14), proving whether
   a failed connection is caused by a broken DNS server or a real network outage.

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